

Exercise 1

Statistical moments. Stationarity tests.

Basic definitions

The operator of expectation (average or mean value of the expression **X**)

$$E[X] = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=0}^N X_i$$

Ordinary moment of k^{th} order

$$m_k = E[X^k]$$

Central moment of k^{th} order

$$\mu_k = E[(X - \bar{X})^k]$$

Statistical measures of signal

Mean value (or average) $\bar{X}$ of random variable **X**

$$\bar{X} = m_1 = E[X]$$

Standard deviation σ

$$\sigma = \sqrt{\mu_2}$$

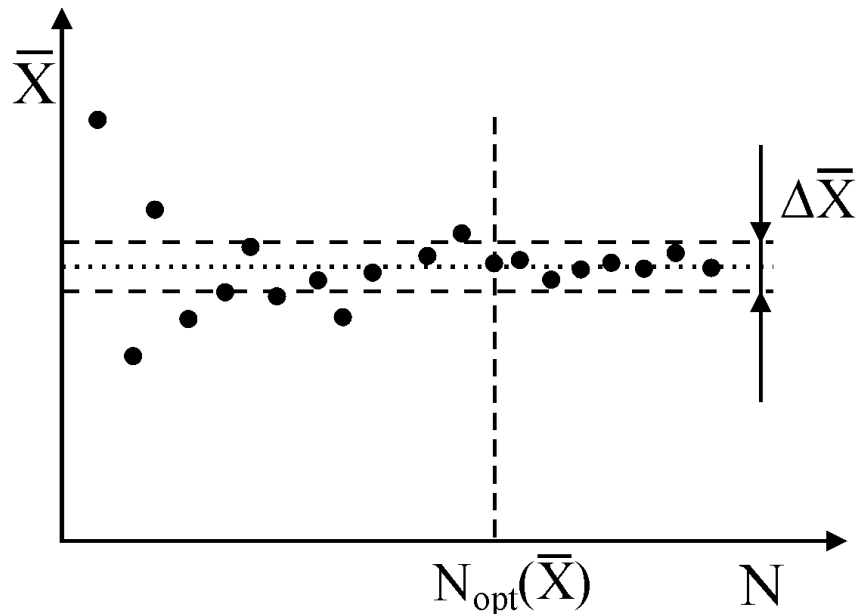
Skewness coefficient **S**

$$S = \frac{\mu_3}{\mu_2^{1.5}} = \frac{\mu_3}{\sigma^3}$$

Flatness coefficient **F**

$$F = \frac{\mu_4}{\mu_2^2} = \frac{\mu_4}{\sigma^4}$$

Stationarity tests



- aim – to check if the signal duration (number of samples) is sufficient to ensure accurate estimation of statistical measures (i.e. mean value $\bar{X}$, standard deviation σ , ...)
- it is necessary to assume the confidence interval (e.g. $\Delta \bar{X}$) for each statistical measure under interest

e.g. $\Delta \bar{X} = 3\% \text{ of } \bar{X}$

e.g. $\Delta \sigma = 5\% \text{ of } \sigma$

- signal duration (number of samples) should "satisfy" all tests, i.e.:
 - $N \geq N_{opt}(\bar{X})$
 - $N \geq N_{opt}(\sigma)$
 -

Compulsory tasks:

1. Stationarity test of selected statistical measure (e.g. mean value) for **periodic** signal (no noise),
2. Stationarity tests of:
 - mean value $\bar{X}$,
 - standard deviation σ ,
 - skewness coefficient S ,
 - flatness coefficient F

for **random** signal. The goal is to find the number of samples N_{opt} ensuring the accurate estimation for the assumed confidence interval,

3. Comparative study of statistical measures for various signal types to find the relationships between signal type and parameters and statistical moments.

Additionally as the **optional** tasks, you can:

4. Determine the influence of a signal variance on $N_{\text{opt}}(\bar{X})$ for the fixed confidence interval,
5. Analyse the dependence of N_{opt} on confidence interval for random signal and selected statistical measures,
6.

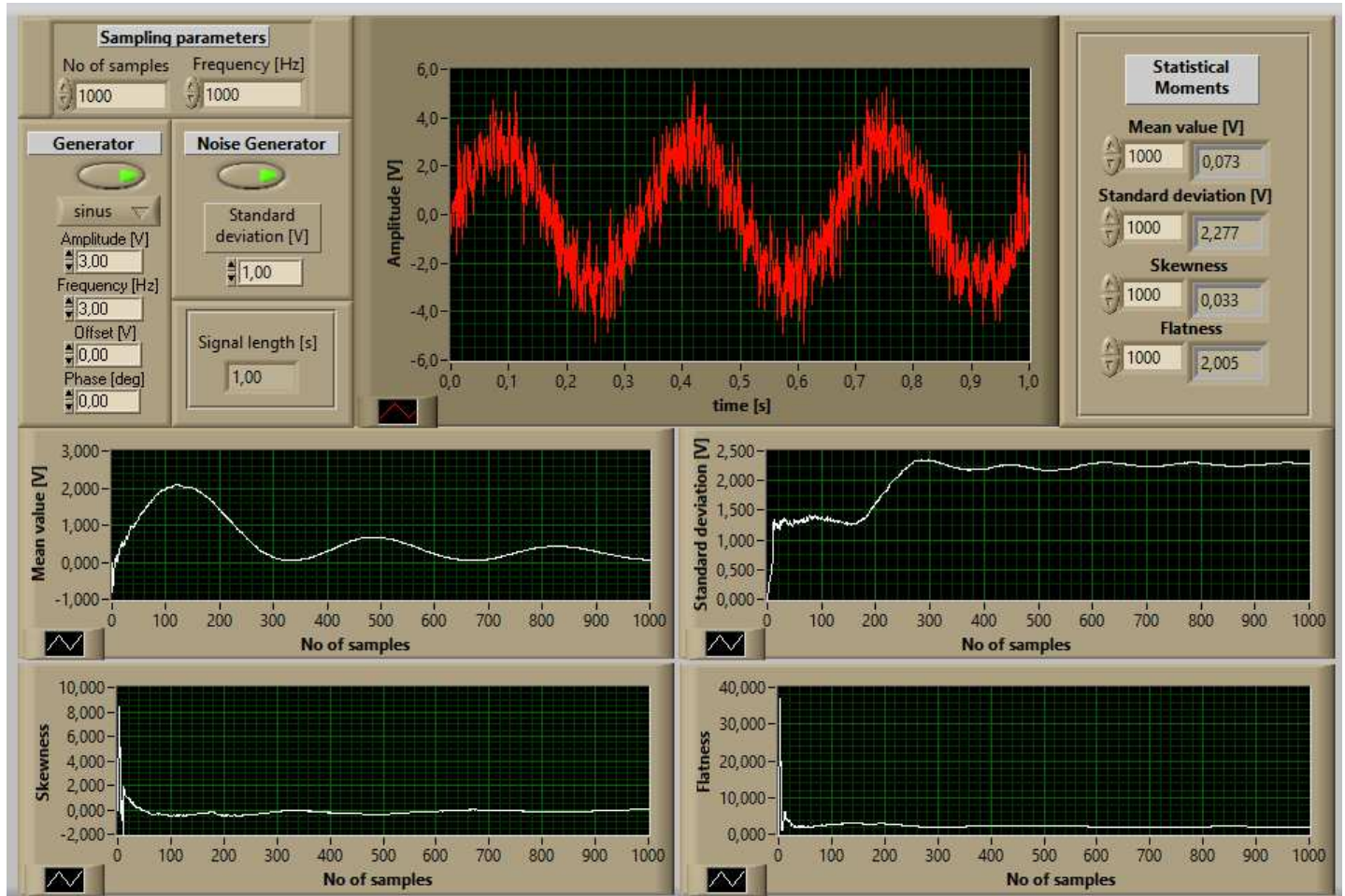


The exercise aims at learning how to:

- **recognise the signal type,**
- **find the signal's parameters**

having known statistical moments (and, of course, not seeing the signal).

LabView environment - project "Moments"



Sample problems

1. When (for what type of signals) there is no need to perform stationarity tests? Please explain.
2. Give the values of the flatness factor for a sine wave and a random (Gaussian) noise.
3. What is the influence of:
 - offset,
 - amplitude,
 - frequency,
 - phase shifton
 - mean value,
 - standard deviation,
 - skewness,
 - flatness.
4. What are the units of variance, skewness and flatness coefficients ?
5. Evaluate the diagnostic value of average, standard deviation, skewness and flatness.
6. What signal parameters are subject to stationarity tests ?
7. What is the relation between the signal's amplitude and its standard deviation for:
 - a) sine wave,
 - b) square wave,
 - c) random noise.
8. What is the physical interpretation of standard deviation ?
9. For the following signals: sine wave and square wave, varying in the same range $2A$ (difference between maximum and minimum values), give the values of their standard deviations.
10. Describe analytically the signal characterised by the following statistical moments:
 - mean value = 3
 - standard deviation = 1.414
 - flatness = 1.5